

3D Printing Zone User Help Manual

When using the Bambu Lab printers, you may encounter issues with either the printer, filament, or AMS feeder. To ensure no issues occur, regular maintenance is highly recommended to minimize interference and maximize efficiency. There may be times where the printer may encounter an issue that regular maintenance cannot fix, and the user might have to either consult a solution in this manual, ask for help, or fix it themselves with their own intuition. When you encounter an issue, please **DO NOT** start taking off parts on the printer if you don't have to, first analyze the problem, and then evaluate your solution. If you need further information, please consult the Bambu labs manual for the appropriate printer and possible fixes provided in this guide.



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Filament Types and Uses

The different filament types we usually have here on campus consist of PLA, PLA+, PETG, and TPU. All these filaments have different properties and can have various uses dependent on the project you have. As a warning, **DO NOT** try to combine these filaments together while printing as they do not bind together in the process.

Here I will list the filaments with their properties, melting temp range, and some examples of usage.

PLA and PLA+

Properties: Stability = Reliable | Print Consistency = Little to no issues | Looks = Clean

Melting Point Print: 205 – 230 °C

Example Projects: Desk Decor, Screws

PETG

Properties: Stability = Weak | Print Consistency = Increased Issues | Looks = Fuzzy/Stringy

Melting Point Print: 230 – 250 °C

Example Projects: Prototyping, Simple Decor

TPU

Properties: Stability = Malleable | Print Consistency = Moderate Issues | Looks = Stringy

Melting Point Print: 190 – 230 °C

Example Projects: Flexible

IMPORTANT NOTE: Print quality varies on how well your printer is set up, make sure it has been maintained recently and that the filament has been dried before use.

Getting Started with Your Printer

When using the printers, there are plenty of similarities and differences between each of them. Luckily, the only one we have that's different is the P1S, which does require a lot of work sometimes to fix. Before starting a print, please make sure the plate is scraped clean of any filament and wiped off with isopropyl alcohol.

A1 and A1 Mini

When using the Bambu Lab A1s or Mini you will be greeted with this home menu. Here are short descriptions of where these pages lead you.

- Print files shows of your current print when its going and the print files on the micro SD card when idle
- Filament displays filament type/status
- Control lets you manually control printer parts when idle
- Settings shows printer information and other maintenance options
- Assistant displays whether the printer is working properly or the error type it has.



Upon entering the settings menu, you will be greeted with a few pages that may list information such as if there is a user account connect, or you may find use in the calibration and firmware update sections. For general use it is recommended that you run the calibration every now and then just keep the printers running appropriately.

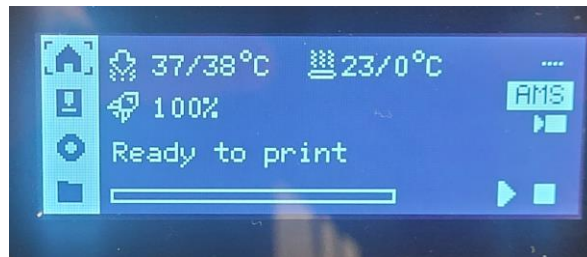


The control menu can be used as follows to switch between the control and XYZ tabs. The XYZ tab is not the most important, but it allows you to mechanically shift your printer bed and extruder along different axis. This can be useful if you want to move parts around to enable better work on certain components during maintenance. The Control tab on the other hand is very important since here you can adjust all your temps and speeds of components along with manually controlling the extruder. Fan speed will help the plastic cool faster, and it is usually left to 100% along with the default speed. It is recommended that if you want a cleaner print with less strings, look at the temp on the filament that is recommended and run it lower manually. For better stick of a print, sometimes turning up the hot bed a few degrees can help. **IMPORTANT NOTE:** If you intend to change values manually, do it after the print starts to print as it will not keep your values when it does pre-calibration processes.

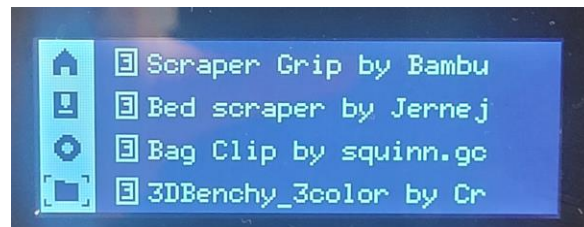


P1S Printer

The P1S printer interface has not been updated by Bambu Labs to properly adapt to newer technologies they have made, which is why using it with the AMS can be a pain. Although for regular use there is a lot less to look at considering the simplicity. The home menu will show stats such as print time, speed, and temps of the extruder and bed. The file menu shown in the middle is where you can manually load files from the micro-SD to the printer. The last menu here is the control menu where you can control printer movements, temps, and speeds manually. To select options in these menus you must navigate to the option via the arrow buttons and then press the middle button to select them.



(Home Menu)



(Files Menu)



(Control Menu)

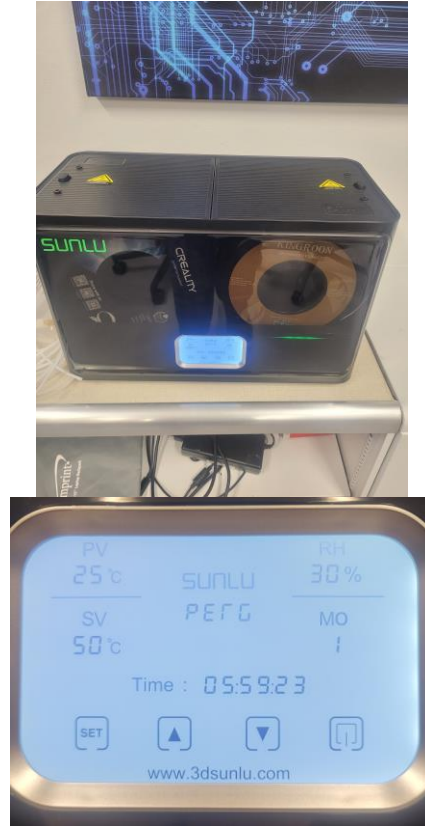
Printing with AMS and Dry Boxes

Both the AMS feeders and dry boxes are essential components to expanding both capabilities and reliability to the 3D printers. They are not difficult to use although you must make a few notes before using them.

The AMS lite and AMS 1.0 are useful for multicolor prints. Please be aware when loading that the enclosed AMS 1.0 will keep the filament dryer for longer periods of time than the AMS lite due to its' enclosure. On the A1/A1 min when you use the AMS lite, the filament menu will now display the AMS tab which is where you can load and unload each spool you need. **DO NOT** Try to mix filament types as mentioned earlier.



It is important to use the dryer boxes consistently on spools to keep moisture out of them. This ensures that prints come out better and more consistent due to the spool breaking less frequently. Use the button on the far right of the display to turn it on, then go and press set until the text in the middle starts flashing. Change the filament type on that flashing text using the arrow buttons and then when the correct filament is selected, please press set repeatedly to set everything else to the correct filament settings. Once you have cycled through all the settings, just leave it to dry.



IMPORTANT NOTICE:

If the filament breaks in the AMS feeder of the P1S, please contact someone who can fix it due to the P1S complexity. If it breaks the A1 or A1 mini, please use the same fix as mentioned on filament stuck fixes (pg. 9-11) to properly fix an A1 printer with the AMS lite.

Filament Stuck Fixes

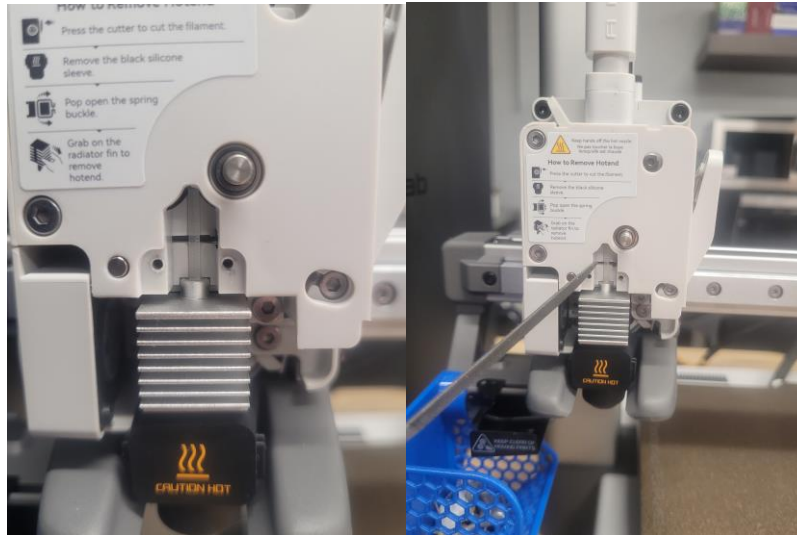
If your filament gets stuck or you need to clean the printer, please use the following instructions for the A1s or the P1S depending on what they apply to.

Filament not Pushing Through (A1/mini)

When the filament is stuck and won't come out one possible way is to do as follows



1. Remove the Front as it just pops off
2. Grab and slip of the yellow and gray circle and then unscrew the 2 pictured screws and remove the plastic bit



3. Once the piece is removed, cut the filament with the handle on the right of the extruder by pushing it in all the way then unload the filament
4. Once the filament is unloaded, set the temp to about 210 °C
5. Use a small thin metal tool of your choice to manually push the filament through the extruder

Once all is done, go to load your filament via the instructions under the filament section and press load to put your filament back in.